MIL-M-21826B

16 February 1973

SUPERSEDING
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28 January 1966

MILITARY SPECIFICATION

MACHINE, PLASTIC SHEET, THERMOFORMING

(FOR PACKAGING APPLICATIONS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the equipment designed to produce plastic skin and blister type packages by vacuum forming methods.

★ 1.2 Classification. Plastic sheet forming machines shall be of the following types, sizes and classes, as specified (see 6.2):

Type I - Manually operated

Size - 15 inch width x 15 inch length
Size - 20 inch width x 20 inch length

Type II - Semiautomatic

Size - 20 inch width x 24 inch length
Size - 20 inch width x 26 inch length

Type III - Class 1 - Automatic

Size - 20 inch width x 24 inch length
Size - 20 inch width x 26 inch length

- Class 2 - Automatic, High Speed

Size - 15 inch width x 12 inch length

FSC 3540

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2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-576	-	Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-B-591	-	Box, Fiberboard, Wood-cleated
PPP-B-601	-	Box, Wood, Cleated-plywood
PPP-B-621	-	Box, Wood, Nailed and Lock-corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-C-650	-	Crates, Wood, Open and Covered

MILITARY

MIL-C-104	-	Crates, Wood, Lumber and Plywood, Sheathed, Nailed and Bolted
MIL-P-116	-	Preservation, Methods of
MIL-E-5558	-	Enamel, Wrinkle-Finish, for Aircraft Use
MIL-P-8585	-	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-E-15090	-	Enamel, Equipment, Light-Gray (Formula No. 111)

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-461	-	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement Of.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Underwriters' Laboratories, Inc.

Standard No. 499 - Electric Heating Appliances

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, Long Island, New York 11746; or 1655 Scott Boulevard, Santa Clara, California 95050.)

National Bureau of Standards

Handbook H28 - Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 So. Riverside Plaza, Chicago, Illinois 60606.)

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

* 3.2 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his commercial product provided that the completed item complies with all provisions of this specification.

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* 3.3 Standard product. The thermoforming plastic sheet machine delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and sub-assemblies thereof, shall be new and interchangeable.

* 3.4 UL standard. Type I, II, and III plastic sheet thermoforming machine heaters shall conform to UL Standard 499.

* 3.4.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the applicable machine heating unit that he proposes to furnish under this specification meets the requirements of UL Standard 499. Acceptable evidence of meeting the requirements of this standard shall be the UL Label or Listing Mark or certification from an independent testing laboratory acceptable to the contracting officer or his authorized representative that the machine heating unit conforms to UL Standard 499.

* 3.5 Design and construction. Type I, II and III, plastic sheet forming machines shall have the following basic components:

(1) A heater to raise the temperature of the plastic sheet to the forming temperature.

(2) A clamping device to contain the plastic sheet while the sheet is heating, forming, and cooling.

(3) A table or platen for placement of the molds for plastic blister shells, or items, for plastic skin-packs.

(4) A vacuum producing system or a means of connection to an external source to evacuate the air between the heated plastic sheet and the blister molds or items.

(5) A frame with mechanical devices to move the clamping device, heater and vacuum table in relation to each other to form the plastic skin-packs or blister forms.

All screw threads used on the machines shall be in accordance with the National Bureau of Standards H-28. Pneumatic systems, when applicable, shall operate between 60 to 80 pounds per square inch (psi).

* 3.5.1 Type I machine. Type I machine operations of feeding, clamping, heater movement, cooling, and removal of the formed plastic sheet shall be performed manually.

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* 3.5.2 Type II machine. Type II machine operations of forming and cooling of the formed plastic sheet shall be automatically performed to occur in sequence. Operations of feeding and removal of the formed sheet shall be performed manually or automatically to occur in sequence with forming and cooling.

* 3.5.3 Type III machine. Type III machine operations of feeding, clamping, heater movement, forming, cooling and removal of the formed plastic sheet shall be automatically performed to occur in sequence.

* 3.5.3.1 Class 1. Type III, class 1 machines shall be capable of completing up to 15 forming cycles per minute with 15-mil cellulose acetate sheets (see 3.7).

* 3.5.3.2 Class 2. Type III, class 2 machines shall have additional preheating means to enable the machine to complete no less than 15 forming cycles per minute with 15-mil cellulose acetate sheets (see 3.7).

* 3.5.4 Size. Size of the type I, II, and III heaters shall consist of the inside area (forming area) of the clamping device (see 1.2).

* 3.5.5 Forming depth. The forming depth of the type I, II and III machines shall consist of the maximum distance between the bottom of the clamping frame and platen. Unless otherwise specified (see 6.2), the forming depth of the type I, II and III machine shall be 3 inches minimum.

* 3.5.6 Heater unit. The heater shall provide uniform heating by means of electrical heating elements enclosed in an insulated sheet metal housing. The heater area shall be positioned not less than 2 inches above the maximum setting of the clamping frame and shall provide uniform radiation of infrared waves to the material. The heating elements shall be arranged in banks to operate between 700-1400°F. and shall be spaced to provide a thermal density of not less than 1.5 kw of power input per square foot of heater area, determined by the ratio of total rated wattage (kilowatts) of heater elements to heater area (square feet). Polished nontarnishing reflectors shall be located over the heating elements to reflect the heat downward in a uniform pattern. Temperature controls, when heating elements provide more than 1100°F temperature, shall be provided to maintain the heater at a constant temperature within ± 10°F, (see 4.7.1). When specified (see 6.2), controls shall be provided that shall furnish full power only when the heater is directly over the forming area (see 3.5.5). The heater shall be designed to permit horizontal positioning the platen, and raising and lowering as required. An insulation box shall completely enclose the heater when the heater is in the non-heating position to prevent heat loss to the working area; to keep electrical consumption to a minimum; and to protect personnel from contact with the heated surface. Movement of the heater unit over the platen shall be accomplished as follows:

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(1) The movement of the heater unit to the heating position in type I machines shall be accomplished manually. The heater handle shall be provided with an insulated shield to prevent accidental contact by the operator with the heated part of the unit.

(2) The movement of the heater unit to the heating position in type II machines shall be accomplished manually or by automatic means. Manually operated heater units shall be provided with an insulated shield to prevent accidental contact by the operator with the heated part. Means shall be provided to increase or decrease the distance between the heater unit and the clamping device. The range of adjustment shall equal the maximum forming depth.

(3) The movement of the heater unit to the heating position in type III machines shall be accomplished by mechanical or compressed air devices automatically actuated by electrical timers and switches. Type III, class 1 machines shall be equipped with means to increase or decrease the distance between the heater unit and the clamping frame. Range of adjustment shall equal the maximum forming depth. Type III, class 2 machines shall be provided with a preheating arrangement so that a high speed production rate of 15 cycles per minute with 15-mil cellulose acetate packages may be attained.

* 3.5.7 Clamping device. Machines shall have a rectangular clamping device to hold the plastic sheet securely during the heating, forming, and cooling operations. The clamping device shall have self-compensating means on all sides for holding plastic sheets ranging in thicknesses from 4 to 125-mils. Forming of the plastic sheet shall be accomplished by lowering the clamping device to the table or platen or by raising the platen to the clamping device. The clamping device for type I, II and III machines shall operate as follows:

(1) Type I, manually.

(2) Type II, manually or automatically after the plastic sheet has been positioned.

(3) Type III, automatically. An interlocking device shall be furnished to prevent the clamping device from opening when the heater is in the plastic heating position.

* 3.5.8 Forming table or platen. The forming table or platen shall be constructed of welded steel or aluminum, reinforced to prevent distortion when the vacuum is applied. The surface of the platen shall be flat and smooth so that airtight seals may be obtained with the base of the mold. The table shall be either stationary or movable in a vertical direction. A vacuum port shall be provided in the table to rapidly evacuate the air beneath the plastic. A uniformly perforated vacuum box for the table shall be provided for skin-packs. The vacuum box shall withstand a vacuum equivalent to 26 inches of mercury with no breakage or visible distortion (see 4.7.2).

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* 3.5.9 Forming operations. Forming operations on type I, II and III machines shall be accomplished by vacuum as follows:

(1) Forming on type I machine shall be accomplished manually or by means of a foot pedal. Control of the vacuum shall be manual.

(2) Forming on type II and III machines shall be accomplished automatically. The forming operations shall be synchronized with the heating cycle and controlled by electrical reset timers.

* 3.5.10 Vacuum system. Type I, II and III machines shall have a vacuum system for the forming operation consisting of a high capacity vacuum pump, a surge tank, valves and piping required to rapidly evacuate the air beneath the formed plastic. The vacuum system shall be leak tight (see 4.7.3).

* 3.5.10.1 Vacuum pump. The vacuum pump shall be of the piston or rotary type and shall have the capacity to permit recovery of the surge tank to a vacuum of 26 inches of mercury for the next operating cycle of the machine. A high vacuum cutoff control shall be furnished to shut-off pump to reduce wear when the maximum vacuum has been obtained.

* 3.5.10.2 Surge tank. The volume of the surge tank for each size machine shall be larger than twice the volume represented by the inside area of the machine clamping device multiplied by the forming depth of the machine (see 3.5.5). The tank shall be made of steel and may be omitted provided that the vacuum pump has sufficient capacity to provide rapid vacuuming and reduced pumpwear as obtainable with the surge tank.

* 3.5.10.3 Piping and valves. Piping and valves of the vacuum system shall be adequate to maintain and control vacuum from the vacuum source. A removable 30 mesh (.002 inch openings) maximum, screen type filter shall be provided for the vacuum port in the table to prevent solid materials from being drawn into the system.

* 3.5.11 Framework. The frame work of type I, II, and III machines shall be steel. Except for open areas required for machine operation, the frame shall be enclosed with removable panels to provide for maintenance, adjustment or inspection of the machine. In addition to the frame, all machines except type I shall have a rack or roller capable of holding a roll of plastic film applicable to machine size, for feeding over the forming table. A rack or roller for type I machine shall be provided when specified (see 6.2).

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* 3.5.12 Forming cycle controls. Forming cycle controls for type I, II and III machines shall consist essentially of controls for operation of the vacuum forming cycle as follows:

(1) Control of the vacuum forming cycle for type I machine shall be accomplished manually.

(2) Control of the vacuum forming cycle for type II and III machines shall be accomplished automatically by mechanical or pneumatic means through such controls as switches, electrical timers, vacuum and pressure gages and pilot lights. The controls shall be conveniently located on a control panel. All operations in the forming cycle shall be synchronized and controlled by electrical reset timers and all phases of the forming cycle shall have an interlocking mechanism between the air and electrical controls to prevent operation of the machine, if air pressure fails.

* 3.5.13 Insulation resistance. Type I, II and III machines shall be designed to operate on 115 volts, single phase, or 220 or 440 volts, 3 phase, alternating current, as specified (see 6.2). All electrical wires shall be adequately insulated and provisions shall be made that such insulation shall not be subject to abrasion from movable parts of the machine. Insulation resistance between current carrying elements and exposed surfaces of the machine shall be 2 megohms when tested as specified in 4.7.4. The main power switch shall be in a readily accessible part of the machine. The electrical circuits for the heater and vacuum pumps shall be activated by toggle switches and shall include fuses or overload relays for protection of the electrical equipment.

* 3.5.14 Accessory operating equipment. Auxiliary equipment consisting of cooling fans, air blow-off system, and combination air regulator shall be provided with type II and III machines.

* 3.5.14.1 Cooling fans. Cooling fans shall be provided and timed to operate after the forming phase of the forming cycle and to cease operation as the clamping frame opens to remove the formed plastic sheets. Fans shall be designed to cool the entire surface of the formed plastic sheet.

* 3.5.14.2 Air blow-off system. An air blow-off system shall be provided in the vacuum table or platen of type II and type III machines to aid in the removal of the formed plastic sheet. The system shall operate by compressed air and shall be equipped with valves to prevent blow-back into the vacuum system. The blow-off system shall have a manual regulator to adjust the pressure. Operation of the blow-off system shall be synchronized to operate automatically with the forming cycle. Switches located on the control panel shall be provided for manual and automatic operation of the blow-off system.

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* 3.5.14.3 Combination air regulator. A combination oiler, air filter and air pressure regulator shall be provided with pneumatic systems, as applicable. The regulator shall be capable of regulating 125 psi maximum inlet air down to the air pressure required for use with air equipment in the machine (see 3.7).

3.6 Finish. All external metal surfaces, except corrosion resistant steel, anodized aluminum, working surfaces, threaded surfaces and cadmium plated parts shall be finished either with one coat of primer conforming to MIL-P-8585 and two coats of light gray semi-gloss enamel conforming to MIL-E-15090, type I, class 2, Hammertone finish or two coats of gray wrinkle conforming to MIL-E-5558, type II. Areas where the heat of the oven will have a deleterious effect upon the paint shall not be painted.

* 3.7 Performance. Type I, II and III machines, as specified in 3.5.1 thru 3.5.3 and in accordance with applicable design and operating characteristics in 3.5.6 thru 3.5.14, shall operate smoothly throughout the forming of 15-mil cellulose acetate film from roll or sheet into well formed multiple blister shells and skin-packs, when tested as specified in 4.7.5.

* 3.8 Electromagnetic compatibility. When specified (see 6.2), type I, II and III machines shall be designed and equipped for electromagnetic compatibility of class IIB equipment in MIL-STD-461 and tested as specified in 4.7.6.

* 3.9 Identification marking. Equipment assemblies and parts shall be marked for identification in accordance with MIL-STD-130. The name plate shall include the following information:

Machine, Plastic Sheet Thermoforming	Forming Depth
Military Specification MIL-M-21826	Manufacturer's Model No.
Type	Manufacturer's Name
Class	Electrical characteristics (see 3.5.13).
Size	

* 3.10 Workmanship. Thermoforming machines, including all parts and accessories, shall be fabricated and finished in accordance with the current commercial practice. Machines shall indicate the following:

(1) Freedom from burrs and sharp edges.

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(2) Thoroughness of soldering, welding, brazing, wiring and riveting.

(3) Alignment and no binding of parts.

(4) Soundness and tightness of screws and bolts.

(5) Smoothness of surface finish, free of runs, embedded foreign matter, tackiness, blisters and areas of no coating of finish coating.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Component and material inspection. The supplier shall insure that all components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards unless excluded, amended, modified, or qualified in this specification.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(1) First Article Inspection (see 4.3).
(2) Quality Conformance Inspection (see 4.4).
(3) Inspection of Preparation for Delivery (see 4.6).

* 4.3 First article. When a first article is required (see 6.2), the machine shall be examined in accordance with table I, dimensions specified and tested in accordance with 4.7.1 through 4.7.6, as applicable. The presence of any defects, dimensional deviations, or failure tc pass any tests shall be cause for rejection of the first article.

* 4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

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4.4.1 End item inspection. The lot shall be all machines of one type, size, and class offered for inspection at one time.

4.4.1.1 Visual examination. The machine shall be examined for the defects in table I, the inspection level shall be II with an acceptable quality level of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Standard product	End item not in accordance with manufacturer's standard product except for changes to comply with specification	X	
Design and construction	Components missing	X	
	Screw threads not in accordance with H-28	X	
	Platen not flat or smooth	X	
	Controls such as switches, electrical timers, vacuum and pressure gages or pilot lights missing on type III machine not conveniently located on control panel	X	
	Design and construction not as specified	X	
Finish	Finish coating not as specified (see 3.6)		X
Marking for identification	Missing, not as specified or not legible	X	
Workmanship	Not free from burrs and sharp edges		X
	Soldering, welding, or brazing not continuous adherent or free from pits, burn-through and flux deposits	X	
	Wiring insecurely joined, broken insulation, frayed or cracked		X
	Riveting not fully peened or loose	X	
	Parts binding or not in alignment	X	

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TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Workmanship (cont'd)	Screws and bolts marked, fractured, stripped, or loose		X
	Finish coating not smooth or free from runs, embedded foreign matter, tackiness, blister or area of no coating		X

* 4.4.1.2 Dimensional examination. The machine shall be examined for compliance with dimensions specified. Any dimension not within specification tolerances shall be a defect. The inspection level shall be S-2 with an AQL of 6.5 defects expressed in terms of defects per hundred units.

* 4.4.1.3 End item testing. When a first article is not required, the first produced machine shall be tested as specified in 4.7.1 through 4.7.6, as applicable. The sample thermoforming plastic sheet machine shall be tested as specified in 4.7.1 through 4.7.5. Failure to pass any test shall be recorded as a defect. The inspection level shall be S-1 with an AQL of 1.5 defects, expressed in terms of defects per hundred units.

* 4.5 Certificate of examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with specification requirements shall be examined and validated.

* 4.6 Inspection of preparation for delivery. Examination will be made to determine that preservation, packing and marking as required by section 5 is complied with. Defects will be scored as specified below. The inspection level shall be S-4 with an acceptable quality level of 6.5, expressed in defects per 100 units.

Examine	Defects
Preservation	Level of preservation, packaging, and packing not as specified (see 5.1 and 5.2).
	Method of preservation, when required, improperly applied or missing.
Container	Boxes not class or style specified.
	Weight limit of shipping container exceeded.

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Examine	Defect
Workmanship	Boxes improperly closed; strapping omitted. Cushioning, blocking and bracing, as required, applied or missing.
Marking	Markings on interior and exterior containers omitted, incorrect or illegible.

4.7 Tests.

* 4.7.1 Uniform temperature. The heater shall be moved over the platen equipped with temperature measuring devices at 5 locations; one at the geometrical center of the platen, and one at each corner, 3 inches from its adjacent sides. The heater shall be turned on and adjusted until a temperature in excess of 1100°F is obtained on the platen. The temperature shall then be taken at the 5 locations to determine compliance with the temperature requirements of 3.5.6. Any nonconformance shall constitute failure to pass this test.

* 4.7.2 Vacuum box strength. With openings sealed, the vacuum box for skin-packs shall be subjected to a vacuum of 26 inches of mercury and examined for compliance with the distortion requirement of 3.5.8. Any nonconformance shall constitute failure to pass this test.

* 4.7.3 Vacuum leak. With vacuum ports in platen sealed, the system vacuum shall be turned "ON" until a vacuum of 26 inches of mercury is obtained. With the vacuum system turned "OFF", the maximum loss in vacuum for a 10-minute holding period shall not exceed 1 inch to determine compliance with 3.5.10. Any nonconformance shall constitute failure to pass this test.

* 4.7.4 Insulation resistance. With all electric switches of the machine turned "ON", a 60-hertz, AC (alternating current) potential of 1000 volts, RMS, plus twice the rated voltage of the machine shall be applied between the exposed surface of the machine as the ground terminal and each of the power input terminals as the positive terminal. The insulation resistance between positive and ground terminals shall then be measured to determine compliance with the 2 megohm requirement of 3.5.13.

* 4.7.5 Performance. The machine shall be operated to form blister shells and skin-packs from the specified plastic film (see 3.7), using items or wooden molds ranging in height to 2 inches, and in length and width to approximately 6 inches. The objects shall be uniformly placed on the vacuum table to allow at least 2 inches and no more than 3 inches between adjacent objects. Machine operation, and the blister shells and skin-packs made shall be examined to determine compliance with 3.7. Any nonconformance shall constitute failure to pass this test.

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* 4.7.6 Electromagnetic compatibility. When electromagnetic compatibility is required, the first article or initial production unit shall be tested by the supplier in accordance with test methods CE03 and RE02 of MIL-STD-462. The Government reserves the right to witness tests performed by the supplier or independent testing agency. The supplier shall furnish the contracting officer written certification that the Interference Control Plan, the EMI/EMC Test Plan, the electromagnetic test report and the requirements meet MIL-STD-461 (see 3.8).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Thermoforming machines shall be preserved and packaged level A or level C as specified (see 6.2).

5.1.1 Disassembly. Disassembly may be required to reduce cubage or for the protection of parts vulnerable to damage. Disassemble to the extent necessary to insure thorough cleaning, drying and preservation provided no special skill or tools are required for reassembly.

5.1.1.1 Level A.

5.1.1.1.1 Cleaning and drying. The machines shall be cleaned in accordance with process C-1 of MIL-P-116 and dried in accordance with procedures D-1 and D-4 of the same specification.

5.1.1.1.2 Unit packaging. Thermoforming machines shall be preserved and packaged in accordance with method IIa of MIL-P-116. Detachable framework or other components not subject to corrosion shall be removed and packaged separately (see 5.1.1). Publications issued with the machine shall be secure to the unit in a conspicious place for the users' convenience. Components removed shall be cushioned and packaged in cleated-paper overlaid veneer, cleated fiberboard, cleated plywood or fiberboard boxes conforming to PPP-B-576, PPP-B-591, PPP-B-601 and PPP-B-636, respectively. Boxes shall be overseas or domestic type as applicable (see 5.2.1 and 5.2.2). The gross weight of wood and wood-cleated boxes shall not exceed the limitations of the applicable specifications. Fiberboard boxes shall not exceed the weight and dimensions of PPP-B-636. Boxes shall be closed and reinforced in accordance with the appendix of the applicable box specification.

5.1.1.2 Level C. Thermoforming machines shall be preserved and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Thermoforming machines shall be packed level A, B or C, as specified (see 6.2).

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5.2.1 Level A. Thermoforming machines, preserved and packaged level A or C as specified, shall be packed in wood cleated plywood boxes, nailed wood boxes, sheathed wood crates or open wood crates conforming to PPP-B-601, overseas type, PPP-B-621, class 2, MIL-C-104, type I and PPP-C-650, style A. Components of the machine detached and packaged in overseas or class 2 boxes specified in 5.1.1.1.2 need not be overpacked. Blocking, bracing and cushioning shall conform to the requirements of MIL-STD-1186. Boxes shall be closed and strapped in accordance with the appendix of the applicable box specification. Unless otherwise specified (see 6.2), the gross weight of boxes shall not exceed 200 pounds unless the weight of a single item exceeds that weight.

5.2.2 Level B. Thermoforming machines, preserved and packaged level A or C as specified, shall be packed in wood cleated paper overlaid veneer, wood cleated plywood boxes, nailed wood boxes or open wood crates conforming to PPP-B-576, class 1, PPP-B-601, domestic type, PPP-B-621, class 1 and PPP-C-650, style B. Components of the machine detached and packaged in boxes specified in 5.1.1.1.2 need not be overpacked. Blocking, bracing and cushioning shall be sufficient to afford adequate protection against damage. Boxes shall be closed in accordance with the appendix of the box specification. Unless otherwise specified, (see 6.2), the gross weight shall not exceed 200 pounds unless the weight of a single item exceeds that weight.

5.2.3 Level C. Machines shall be packed in exterior shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall conform to Uniform Freight Classification rules or National Motor Freight Classification rules, as applicable.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The thermoforming machines covered by this specification are intended for use in forming skin type and blister type packages for the preservation of small parts and components.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.3 and 6.3).
(c) Type, class and size of machine required (see 1.2).

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(d) When machine forming depth other than specified is required (see 3.5.5).
(e) When controls for full heater power over forming area are required (see 3.5.6).
(f) When rack or roller for type I machine is required (see 3.5.11).
(g) Machine electrical characteristics required (see 3.5.13).
(h) When electromagnetic compatibility is required (see 3.8).
(i) Required level of preservation, packaging and packing (see 5.1 and 5.2).
(j) When container weight other than specified is required (see 5.2.1 and 5.2.2).

6.2.1 Contract data requirements. Any requirement for equipment manuals for the items covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a First Article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire contents irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 84

Preparing activity:

Army - GL

Project No. 3540-0106

Review activity:

Army - SM

User activities:

Navy - MC, SH

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Machine, Plastic Sheet, Thermoforming (for Packaging Applications) MIL-M-21826B

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

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